

EM-II
Practice Question Sheet #1
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Topics: Vector Calculus

1. Find the gradient of the scalar field $f(x, y) = y^3 + 4xy$.
2. Give the geometrical interpretation of gradient. Find the unit normal vector to the surface $x^2 + y^2 + z^2 = 4$ at the point $(\frac{1}{\sqrt{2}}, \sqrt{3}, \frac{1}{\sqrt{2}})$.
3. Find the directional derivative of $f(x, y, z) = xy^2 \sin z + 4y^2z + 2z^2$ at the point $(1, -1, 2)$ in the direction of $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$.
4. Find the directional derivative of $f(x, y, z) = xy^2 + 4y^2z + 2z^2$ at the point $P(1, -1, 2)$ in the direction of PQ , where $Q(0, 1, -1)$.
5. Find the directional derivative of $f(x, y, z) = xy + yz + zx$ at the point $(0, -1, 1)$ in the direction of $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.
6. Find the divergence of the vector field $\mathbf{v} = (x^2y^2)\mathbf{i} + xyz\mathbf{j} - \sin zx\mathbf{k}$.
7. Show that the vector field $\mathbf{F} = (2x + 3y)\mathbf{i} + (x - y)\mathbf{j} - (x + y + z)\mathbf{k}$ is incompressible/solenoidal.
8. Show that the vector field $\mathbf{F} = yz^2\mathbf{i} + xz^2\mathbf{j} + 2xyz\mathbf{k}$ is conservative. Find its potential function.
9. If r is the distance of a point (x, y, z) from the origin, prove that $\text{Curl}\left(k \times \text{grad} \frac{1}{r}\right) = -\text{grad}\left(k \cdot \text{grad} \frac{1}{r}\right)$, where k is the unit vector in the direction of OZ .
10. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = x^2\mathbf{i} + xy\mathbf{j}$ and C is the boundary of the square in the plane $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$ and $y = a$.
11. Evaluate $\int_C (xy)dx - x^2dy + (yz)dz$, where C is the curve defined by $y = x^2$, $z = 2x$, $-1 \leq x \leq 3$.
12. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ where $\mathbf{F} = z^2y\mathbf{i} + x^2y\mathbf{j} - y^2z\mathbf{k}$ and S is the portion of the surface of the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 3$ included in the first octant.
13. Use the Gauss theorem to evaluate $\iint_S (\mathbf{V} \cdot \mathbf{n})dA$, where $\mathbf{V} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$ and S is the boundary of ellipsoid $2x^2 + 3y^2 + z^2 = 4$.
14. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using the Stokes's theorem where $\mathbf{F} = y^2\mathbf{i} + x^2\mathbf{j} - (x + z)\mathbf{k}$ and C is the boundary of triangle with vertices at $(0, 0, 0)$, $(1, 0, 0)$ and $(1, 1, 0)$.
15. Evaluate $\oint_C \mathbf{V} \cdot d\mathbf{r}$ using the Stokes's theorem where $\mathbf{V} = 3ye^x\mathbf{i} + 4z \sin y\mathbf{j} + 2x \cos z\mathbf{k}$ and C is the intersection of the surface of the sphere $x^2 + y^2 + z^2 = 4$, $x \geq 0$ and the cylinder $y^2 + z^2 = 1$.